

Generative AI Declaration

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Summary

In accordance with Monash University's Academic Integrity Policy and the FIT5226 assignment specifications, I declare the following use of Generative AI tools during the completion of Project Stage 2. **NotebookML** was used for understanding and breaking down the project. **Claude** was used primarily for code assistance and mathematical formatting. **Gemini** was used primarily for document parsing and analytical proofreading. All use is documented below with representative prompts. The final submitted work represents my own understanding, design decisions, and synthesis of the assignment requirements.

1. Understanding and Breaking Down the Project

Tool used: NotebookML

Nature of use: Passive, parsing and summarising existing documents only. No content was generated.

AI was used at the outset to parse the assignment specification and map it against relevant course materials, such as lectures on MARL and Game Theory. This ensured a correct understanding of the scope of each task before implementation began.

Representative prompts:

"Act as an expert Multi-Agent Reinforcement Learning (MARL) professor. Map the requirements of the FIT5226 Stage 2 project against course materials including MDPs, Game Theory solution concepts, and the implementation patterns for Tabular Q-Learning."

"Read the staff clarifications provided in the Ed Forum regarding robot streams, shared policies per type, and collision resolution. Summarize the constraints for environment step resolution."

What I retained responsibility for: Interpreting the specifications, making all core architectural design decisions, and synthesising staff clarifications into a cohesive simulation logic. The AI produced summaries; all decisions following those summaries were my own.

2. Code Assistance and Refactoring

Tool used: Claude

Nature of use: Active, AI generated initial drafts of specific modules, which were subsequently reviewed, tested, and modified by me.

AI functioned as a pair-programmer for the following specific modules. All generated code was reviewed against the assignment constraints, particularly the "no heuristics in reward schema" rule, before being integrated.

Areas of assistance:

- **Configuration Management:** AI generated initial drafts of Python dataclasses (`AgentConfig` , `EnvConfig`) and JSON structures for hyperparameter management, including the discount factor γ and step penalties. I revised these to match the environment specifications.
- **Code Quality:** AI refactored variable names and docstrings for readability. The underlying logic was written by me and was not altered during refactoring.
- **Environment Logic:** AI provided a structural template for the simultaneous step resolution module. I implemented the final ordering logic, including resolving lake states before agent moves and the collision detection rules.
- **Visualization:** AI assisted with library API calls for policy quiver plots and Phase 2 replicator dynamics phase portraits. I specified the data structures and verified the visual outputs against expected behaviour.

What I retained responsibility for: The correctness of the tabular Q-learning algorithm, the specific ordering of all environment transitions, reward schema design, and all debugging and testing.

3. Mathematical Formulation and LaTeX

Tool used: Claude
Nature of use: Passive, formatting only. All mathematical content was derived independently before AI assistance was sought.

AI was used exclusively to format pre-derived mathematical expressions into LaTeX for the Jupyter Notebook documentation. The derivations, values, and logic were completed by me prior to any AI involvement.

Areas of assistance:

- Formatting the **Bellman Expectation Equation** to account for lake stochasticity, based on Sarah's clarification.
- Formatting the **Replicator Dynamics ODEs** and payoff matrices for the game-theoretic analysis in Task 3.

Representative prompt:

"Format the following hand-derived equation into LaTeX, left-aligned, without altering any values or structure: [equation]."

What I retained responsibility for: The derivation of all payoff values, calculation of Nash Equilibria for the reduced crossing game, stability analysis of fixed points, and verification that the formatted output matched my original derivations exactly.

4. Analytical Writing and Proofreading

Tool used: Gemini
Nature of use: Passive, proofreading only. No analytical content was generated or suggested by the AI.

AI was used solely as a proofreading tool to check grammatical accuracy and ensure the analytical sections of Tasks 3 and 4 remained within the strict 150-word limit. The AI was explicitly instructed not to suggest content changes.

Areas of assistance:

- Checking clarity and conciseness of the argument regarding why Phase 2 is harder for independent learners due to non-stationarity and the moving target problem.
- Reviewing the conceptual link between reinforcement learning outcomes and evolutionary game theory for grammatical accuracy.

Representative prompt:

"Proofread the following paragraph for grammar and word count only. Do not suggest changes to the argument, structure, or technical content: [paragraph]."

What I retained responsibility for: All theoretical arguments, interpretations of simulation data, and the final synthesis of findings. The submitted analytical text is entirely student-authored.

Declaration of Understanding

I confirm that I have reviewed, tested, and fully understand all code and theoretical arguments presented in this submission. I am prepared to independently explain the following without AI assistance:

- **Task 1 and 2 — Environment Logic:** Why a collision is registered only at the final result of simultaneous steps, and the specific ordering of lake state resolution before agent movement.
- **Task 1 and 2 — Learning Algorithm:** The necessity of accounting for both future lake states in the TD-target calculation, and how the Q-table update rule was implemented.
- **Task 3 — Game Theory:** The derivation of payoff values, the calculation of Nash Equilibria for the reduced crossing game, and the stability analysis of replicator dynamics fixed points.

- **Task 3 and 4 — Comparative Analysis:** The convergence of Phase 1 due to dominant strategies versus the coordination failure in Phase 2 caused by non-stationarity, and how the replicator dynamics simulation mirrors Q-agent learning behaviour.

The final submitted work represents my own understanding and synthesis of the assignment requirements.